Ken Yang

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TECHNICAL SKILLS

Programming & Development: Java, Python, JavaScript, C++, HTML & CSS
Testing: JUnit, JaCoCo, Integration Testing, Manual Testing
Tools & Platforms: Git, GitHub, VS Code, IntelliJ IDEA
Productivity & Media: Microsoft Office Suite (Excel, PowerPoint, Word), CapCut, Canva
Language Proficiencies: English & Chinese (Mandarin)

PROJECTS

CMPT 276 – 2D Java Game (Zombie Run) | Java, Java Swing, JUnit, JaCoCo **Apr 2026**

- **Architected** the class diagram and led system design, establishing modular structure across core game components following software engineering principles
- **Engineered** player movement, enemy BFS pathfinding, and wall collision detection in Java, delivering reliable and responsive gameplay across all levels
- **Documented system architecture and naming conventions to maintain clarity across the codebase.**

GDSC Hack the Sem – EMBER | BC Wildfire Evacuation Platform | Won 1st Place **Apr 2026**

- **Contributed** to quality assurance through testing, helping validate briefing accuracy and evacuation logic across demo scenarios
- **Collaborated** in an Agile hackathon team to deliver a 1st place winning full-stack evacuation intelligence platform built with React, FastAPI
- Organized geospatial, shelter, and road-closure data in Excel and created the project's PowerPoint presentation.

CMPT 225 – Rubik's Cube Solver | IDA* | Java **Dec 2025**

- **Built** an optimal solver using cubie-based representation + IDA* with custom heuristics to prune search space
- **Optimized** solver performance by precomputing move tables and minimizing state transitions enabling the solver to complete complex scrambles within strict runtime limits.
- **Ensured** algorithm correctness through systematic state mapping validation and debugging tests preventing invalid cube states and incorrect solution paths.

VOLUNTEER

Calc Connect Peer **May 2026 - Present**

- **Facilitated** collaborative calculus problem-solving for 30-student MATH 150 seminars by guiding groups through whiteboard exercises and breaking down complex concepts into clear, actionable steps.
- **Strengthened** student performance by mentoring first-year learners on study strategies, resource use, and effective approaches to university-level math.
- **Supported** instructional delivery by coordinating with course instructors to set up seminar activities, prepare problem sets, and maintain an engaging, supportive learning environment.

EDUCATION

Simon Fraser University - Bachelor of Computing Science **January 2025 - Present**
Kwantlen Polytechnic University - Bachelor of Information Technology **Sep 2023 - Dec 2024**
Dean's Honour Roll, Kwantlen Polytechnic University (2024)